

# Elementary Problems

## Triangle Geometry

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## Basic Work

1. (Ukraine MO 2009 Grade 8 Problem 6) In acute-angled triangle  $ABC$ , let  $M$  be the midpoint of  $BC$  and let  $K$  be a point on side  $AB$ . We know that  $AM$  meets  $CK$  at  $F$ . Prove that if  $AK = KF$  then  $AB = CF$ .
2. Let  $W, X, Y, Z$  be the midpoints of the sides  $AB, BC, CD, DA$  of a convex quadrilateral  $ABCD$  respectively. Prove that  $WXYZ$  is a parallelogram.
3. In  $\triangle ABC$ , the internal angle bisector of  $\angle BAC$  meets the side  $BC$  at  $E$ , and meets the circumcircle of  $\triangle ABC$  again at  $D$ . Let  $M$  and  $N$  be the midpoints of  $BD$  and  $AD$  respectively. Prove that  $B, M, E, N$  are concyclic.
4. In  $\triangle ABC$  with  $AB < AC$ , point  $D$  is the foot of the internal angle bisector of  $\angle BAC$ . Let  $E$  and  $F$  be points lying on the sides  $AC$  and  $AB$  respectively such that  $DE \parallel BA$  and  $DF \parallel CA$ . The extension of  $EF$  and  $CB$  meet at  $P$ . Prove that  $PA$  is tangent to the circumcircle of  $\triangle ABC$ .
5. In  $\triangle ABC$ , we have  $AB + AC = 2BC$ . Let  $D$  be the foot of the internal angle bisector from  $A$ , and let  $M$  be the midpoint of  $AB$ . Prove that  $BD = BM$ .
6. (Iranian Geometry Olympiad 2015 Elementary)  
Let  $ABC$  be a triangle with  $\angle A = 60^\circ$ . The points  $M, N, K$  lie on  $BC, AC, AB$  respectively such that  $BK = KM = MN = NC$ . If  $AN = 2AK$ , find the values of  $\angle B$  and  $\angle C$ .
7. (Iranian Geometry Olympiad 2014 Junior) In a right triangle  $ABC$  we have  $\angle A = 90^\circ$ ,  $\angle C = 30^\circ$ . Denote by  $\mathcal{C}$  the circle passing through  $A$  which is tangent to  $BC$  at the midpoint. Assume that  $\mathcal{C}$  intersects  $AC$  and the circumcircle of  $ABC$  at  $N$  and  $M$  respectively. Prove that  $MN \perp BC$ .
8. (IMO shortlist 1969) A right-angled triangle  $OAB$  has its right angle at the point  $B$ . An arbitrary circle with centre on the line  $OB$  is tangent to the line  $OA$ . Let  $AT$  be the tangent to the circle different from  $OA$  ( $T$  is the point of tangency). Prove that the median from  $B$  of the triangle  $OAB$  intersects  $AT$  at a point  $M$  such that  $MB = MT$ .

## Centroid

9. (IMO shortlist 1966) Let  $ABCD$  be a quadrilateral inscribed in a circle. Show that the centroids of triangles  $ABC, CDA, BCD, DAB$  lie on one circle.

10. Let  $G$  be the centroid of  $\triangle ABC$ . Let  $P$  be the foot of the altitude from  $A$ . The line through  $A$  parallel to  $BC$  meets the circumcircle of  $\triangle ABC$  again at  $D$ . Prove that  $P, G, D$  are collinear.
11. (Tournament of Towns 2009) Let  $ABCD$  be a rhombus.  $P$  is a point on side  $BC$  and  $Q$  is a point on side  $CD$  such that  $BP = CQ$ . Prove that centroid of triangle  $APQ$  lies on the segment  $BD$ .
12. (IMO shortlist 1969) Given triangle  $ABC$ , points  $M$  and  $N$  are in the sides  $AB$  and  $AC$  respectively.  
If  $\frac{BM}{MA} + \frac{CN}{NA} = 1$ , then prove that the centroid of  $ABC$  lies on  $MN$ .

### Circumcentre

13. Let  $O$  be the circumcentre of an acute  $\triangle ABC$ . The perpendicular bisector of  $AB$  meets the side  $AC$  at  $D$ . The perpendicular bisector of  $AC$  meets the side  $AB$  at  $E$ . Prove that  $E, B, C, D, O$  are concyclic.
14. In  $\triangle ABC$ ,  $D$  and  $E$  are points on the side  $BC$  such that  $\angle BAD = \angle CAE$ . Let  $O$  be the circumcentre of  $\triangle ABC$  and  $O'$  the circumcentre of  $\triangle ADE$ . Prove that  $A, O, O'$  are collinear.
15. Let  $O$  be the circumcentre of  $\triangle ABC$ . The tangent at  $A$  to the circumcircle of  $\triangle ABC$  meets the line through  $C$  perpendicular to  $BC$  at  $D$ . Let  $E$  be the point on  $BC$  such that  $DE \parallel AB$ . Prove that  $A, O, E$  are collinear.
16. Let  $AB$  be a diameter of a circle  $\Gamma$ . Point  $C$  lies outside  $\Gamma$ . The lines  $AC$  and  $BC$  meet  $\Gamma$  again at  $P$  and  $Q$  respectively. Suppose  $P$  lies between  $A$  and  $C$ , and  $B$  lies between  $C$  and  $Q$ . Let  $O$  be the circumcentre of  $\triangle CPQ$ . Prove that  $OP$  and  $OQ$  are tangents to  $\Gamma$ .
17. (APMO 2010 Problem 1) Let  $ABC$  be a triangle with  $\angle BAC \neq 90^\circ$ . Let  $O$  be the circumcentre of the triangle  $ABC$  and let  $\Gamma$  be the circumcircle of the triangle  $BOC$ . Suppose that  $\Gamma$  intersects the line segment  $AB$  at  $P$  different from  $B$ , and the line segment  $AC$  at  $Q$  different from  $C$ . Let  $ON$  be a diameter of the circle  $\Gamma$ . Prove that the quadrilateral  $APNQ$  is a parallelogram.
18. Let  $D$  be the point on the circumcircle of  $\triangle ABC$  such that  $AD \perp BC$ . Let  $E$  be a point on the arc  $AC$  not containing  $B$ . The line  $DE$  meets  $BC$  at  $F$ . Prove that the circumcentre of  $\triangle BFE$  lies on  $AB$ .
19. (Mexico MO 2009 Problem 1) In  $\triangle ABC$ , let  $D$  be the foot of the altitude from  $A$  to  $BC$ . A circle centred at  $D$  with radius  $AD$  intersects lines  $AB$  and  $AC$  at  $P$  and  $Q$ , respectively. Show that  $\triangle AQP \sim \triangle ABC$ .

20. In  $\triangle ABC$ , the altitude from  $A$  meets the circumcircle again at  $D$ . Let  $O$  be the circumcentre of  $\triangle ABC$ . The lines  $OA$  and  $OD$  meet the side  $BC$  at  $E$  and  $F$  respectively. Prove that  $\triangle CFD \sim \triangle AEC$ .
21. Let  $D$  be a point on the side  $BC$  of  $\triangle ABC$ . Let  $O, X, Y$  be the circumcentres of  $\triangle ABC, \triangle ABD, \triangle ACD$  respectively. Prove that  $A, X, O, Y$  are concyclic.
22. (Iranian Geometry Olympiad 2015 Medium) In acute-angled triangle  $ABC$ ,  $BH$  is the altitude of the vertex  $B$ . The points  $D$  and  $E$  are midpoints of  $AB$  and  $AC$  respectively. Suppose that  $F$  is the reflection of  $H$  with respect to  $ED$ . Prove that the line  $BF$  passes through circumcentre of  $ABC$ .
23. (USAMO 1993 Problem 2) Let  $ABCD$  be a convex quadrilateral such that diagonals  $AC$  and  $BD$  intersect at right angles, and let  $E$  be their intersection. Prove that the reflections of  $E$  across  $AB, BC, CD, DA$  are concyclic.
24. (Poland MO 2nd Round 2009 Problem 4)  $ABCD$  is a cyclic quadrilateral inscribed in the circle  $\Gamma$  with  $AB$  a diameter. Let  $E$  be the intersection of the diagonals  $AC$  and  $BD$ . The tangents to  $\Gamma$  at the points  $C, D$  meet at  $P$ . Prove that  $PC = PE$ .
25. (Russian MO 2000 Grade 9 Problem 3) Let  $O$  be the centre of the circumcircle  $\omega$  of an acute-angled triangle  $ABC$ . A circle  $\omega_1$  with centre  $K$  passes through  $A, O, C$  and intersects  $AB$  at  $M$  and  $BC$  at  $N$ . Point  $L$  is symmetric to  $K$  with respect to line  $NM$ . Prove that  $BL \perp AC$ .
26. Let  $A, B, C$  be three points on a line in that order. The squares  $ABDE$  and  $BCPQ$  are constructed on the same side of the line  $AC$ . Let  $X$  be the point on the segment  $AB$  such that  $AX = BC$ . Prove that  $E, X, B, P$  are concyclic.

### Orthocentre and Nine Point Circle

27. Let  $ABCD$  be a concave quadrilateral such that  $\angle BAD = \angle CBA = \angle ADC = 45^\circ$ . Prove that  $AC = BD$ .
28. Let  $D, E, F$  be the feet of the altitudes from  $A, B, C$  of  $\triangle ABC$  respectively. Let  $H$  be the orthocentre of  $\triangle ABC$ . Prove that  $H$  is the incentre of  $\triangle DEF$ .
29. (IMO shortlist 1988) Let  $ABC$  be an acute-angled triangle. The lines  $L_A, L_B$  and  $L_C$  are constructed through the vertices  $A, B$  and  $C$  respectively according the following prescription: Let  $H$  be the foot of the altitude drawn from the vertex  $A$  to the side  $BC$ ; let  $S_A$  be the circle with diameter  $AH$ ; let  $S_A$  meet the sides  $AB$  and  $AC$  at  $M$  and  $N$  respectively, where  $M$  and  $N$  are distinct from  $A$ ; then let  $L_A$  be the line through  $A$  perpendicular to  $MN$ . The lines  $L_B$  and  $L_C$  are constructed similarly. Prove that the lines  $L_A, L_B$  and  $L_C$  are concurrent.

30. Let  $H$  be the orthocentre of acute  $\triangle ABC$ . Let  $P$  be a point inside  $\triangle ABC$ . The line obtained by reflecting  $AP$  in the line  $AB$  meets the circumcircle of  $\triangle AHB$  again at  $D$ . The line obtained by reflecting  $AP$  in the line  $AC$  meets the circumcircle of  $\triangle AHC$  again at  $E$ . Prove that  $D, H, E$  are collinear.
31. (Mexico MO 2011 Problem 2) Let  $ABC$  be an acute triangle and  $\Gamma$  its circumcircle. Let  $\ell$  be the line tangent to  $\Gamma$  at  $A$ . Let  $D$  and  $E$  be the intersections of the circle with centre  $B$  and radius  $AB$  with lines  $\ell$  and  $AC$ , respectively. Prove that orthocentre of  $ABC$  lies on line  $DE$ .
32. (Philippines MO 2011 Problem 2) In triangle  $ABC$ , let  $X$  and  $Y$  be the midpoints of  $AB$  and  $AC$ , respectively. On segment  $BC$ , there is a point  $D$ , different from its midpoint, such that  $\angle XDY = \angle BAC$ . Prove that  $AD \perp BC$ .
33. Let  $D, E, F$  be the feet of the altitudes from  $A, B, C$  of  $\triangle ABC$  respectively. Let  $P$  be the intersection of  $AD$  and  $EF$ . Prove that the circumcircles of  $\triangle BCE$  and  $\triangle DEP$  are tangent to each other.
34. In  $\triangle ABC$ ,  $\angle BAC = 90^\circ$ . Let  $D$  be the foot of the altitude from  $A$ , and let  $E$  be the midpoint of  $CD$ . Let  $F$  be the reflection of  $A$  in the point  $B$ . Prove that  $FD \perp AE$ .
35. Let  $BD$  and  $CE$  be the altitudes of an acute  $\triangle ABC$ , with  $D$  and  $E$  lying on the sides  $AC$  and  $AB$  respectively. Let  $M$  be the midpoint of  $BC$ . The line  $AM$  meets the circumcircle of  $\triangle ABC$  again at  $P$ . The point  $Q$  lies on the segment  $AM$  such that  $MQ = MP$ . Prove that  $A, E, Q, D$  are concyclic.
36. Let  $H$  be the orthocentre of an acute  $\triangle ABC$ . Let  $P$  be any point on the side  $BC$ . The perpendicular line through  $P$  to  $BC$  meets the lines  $AC$  and  $AB$  at  $D$  and  $E$  respectively. Prove that the orthocentre  $H'$  of  $\triangle ADE$  lies on the line  $HP$ .
37. Let  $H$  be the orthocentre of  $\triangle ABC$ . Let  $P, D, E$  be the midpoints of  $AH, BC, CA$  respectively. Prove that  $\angle PED = 90^\circ$ .
38. (Czech and Slovak Olympiad Round III Category A 2011 Problem 5) In acute triangle  $ABC$ , which is not equilateral, let  $P$  denote the foot of the altitude from  $C$  to side  $AB$ ; let  $H$  denote the orthocentre; let  $O$  denote the circumcentre; let  $D$  denote the intersection of line  $CO$  with  $AB$ ; and let  $E$  denote the midpoint of  $CD$ . Prove that line  $EP$  passes through the midpoint of  $OH$ .
39. (IMO shortlist 1966) Given four points  $A, B, C, D$  on a circle such that  $AB$  is a diameter and  $CD$  is not a diameter, show that the line joining the point of intersection of the tangents to the circle at the points  $C$  and  $D$  with the point of intersection of the lines  $AC$  and  $BD$  is perpendicular to the line  $AB$ .

## Incentre and Excentres

40. In  $\triangle ABC$ ,  $\angle BAC = 60^\circ$ . Let  $O, H, I$  be the circumcentre, orthocentre and incentre of  $\triangle ABC$ . Prove that  $B, C, O, I, H$  are concyclic.
41. Let  $ABCD$  be a cyclic quadrilateral. Let  $I_1$  and  $I_2$  be the incentres of  $\triangle ABC$  and  $\triangle DBC$  respectively. Prove that  $B, C, I_2, I_1$  are concyclic.
42. (USAMO 1988 Problem 4) Let  $I$  be the incentre of triangle  $ABC$ , and let  $A', B'$ , and  $C'$  be the circumcentres of triangles  $IBC$ ,  $ICA$ , and  $IAB$ , respectively. Prove that the circumcircles of triangles  $ABC$  and  $A'B'C'$  are concentric.
43. (CGMO 2012 Problem 5)  
The incircle  $\odot I$  of  $\triangle ABC$  is tangent to the sides  $AB, AC$  at points  $D, E$  respectively. The circumcentre of  $\triangle BCI$  is  $O$ . Prove that  $\angle ODB = \angle OEC$ .
44. In  $\triangle ABC$ , point  $D$  lies on the side  $BC$ . Points  $E$  and  $F$  lie on the sides  $AB$  and  $AC$  respectively such that  $DE = DB$  and  $DF = DC$ . Let  $I$  be the incentre of  $\triangle DEF$ . Prove that  $A, E, I, F$  are concyclic.
45. Let  $I$  be the incentre of  $\triangle ABC$ . The incircle  $\Gamma$  of  $\triangle ABC$  touches the sides  $BC, CA, AB$  at points  $D, E, F$  respectively. The segments  $AI, BI, CI$  meet  $\Gamma$  at  $X, Y, Z$  respectively. Prove that  $DX, EY, FZ$  are concurrent.
46. (Belarus MO 2012 Problem 6) Let point  $I$  be the incentre of  $\triangle ABC$ . Ray  $AI$  intersects circumcircle of  $\triangle ABC$  at point  $D$ . Circumcircle of  $\triangle CDI$  intersects ray  $BI$  at points  $I$  and  $K$ . Prove that  $BK = CK$ .
47. (Russian MO 1996 Grade 9 Problem 6) In the isosceles triangle  $ABC$  ( $AC = BC$ ) point  $O$  is the circumcentre,  $I$  the incentre, and  $D$  lies on  $BC$  so that lines  $OD$  and  $BI$  are perpendicular. Prove that  $ID$  and  $AC$  are parallel.
48. (Poland MO 2nd Round 2013 Problem 2) Circles  $\omega_1$  and  $\omega_2$  with centres in  $O_1$  and  $O_2$ , respectively, intersect in two different points  $A$  and  $B$ , wherein angle  $O_1AO_2$  is obtuse. Line  $O_1B$  intersects circle  $\omega_2$  in point  $C \neq B$ . Line  $O_2B$  intersects circle  $\omega_1$  in point  $D \neq B$ . Show that point  $B$  is incentre of triangle  $ACD$ .
49. (Bosnia Herzegovina TST 2011 Problem 1) In triangle  $ABC$  it holds  $BC = \frac{1}{2}(AB+AC)$ . Let  $M$  and  $N$  be midpoints of  $AB$  and  $AC$ , and let  $I$  be the incentre of  $ABC$ . Prove that  $A, M, I, N$  are concyclic.
50. (Sharygin Geometry Olympiad 2009) Given triangle  $ABC$ , point  $O$  is the centre of the excircle touching the side  $BC$ . Point  $O_1$  is the reflection of  $O$  in  $BC$ . Determine angle  $A$  if  $O_1$  lies on the circumcircle of  $ABC$ .

51. (Bosnia Herzegovina TST 2009 Problem 1) Denote by  $M$  and  $N$  feet of perpendiculars from  $A$  to angle bisectors of exterior angles at  $B$  and  $C$  in  $\triangle ABC$ . Prove that the length of segment  $MN$  is equal to semiperimeter of  $\triangle ABC$ .
52. (Iranian Geometry Olympiad 2017 Advanced) In triangle  $ABC$ , the incircle, with centre  $I$ , touches the side  $BC$  at point  $D$ . Line  $DI$  meets  $AC$  at  $X$ . The tangent line from  $X$  to the incircle (different from  $AC$ ) intersects  $AB$  at  $Y$ . If  $YI$  and  $BC$  intersect at point  $Z$ , prove that  $AB = BZ$ .
53. Let  $ABCD$  be a trapezoid with  $AD \parallel BC$ . Let  $I$  be the incentre of  $\triangle ABC$  and  $E$  the  $A$ -excentre of  $\triangle ACD$ . Suppose  $IE \parallel AD$ . Prove that  $AICE$  and  $ABCD$  are isosceles trapezoid.

## Record

v1.0: problems 1–53